

1. **Experiment:** Computational design strategy
2. **Time:** 2025.10.26-2025.11.04
3. **Member:** Zheyuan Deng, Kaili Jin, Ziyang Yu, Qichang Liu, Yulongzi Piao, Haochen Xu
4. **Method:**

(1) Structure acquisition of the IL-6/sIL-6Rα/gp130 ternary complex

① The X-ray crystal structure of the human IL-6/sIL-6Rα/gp130 ternary signaling complex (PDB ID: 1P9M) was retrieved from the Protein Data Bank (PDB) and imported into PyMOL for structural inspection.

② IL-6, sIL-6Rα, and gp130 were displayed as separate molecular components in PyMOL to enable direct visualization of their relative spatial arrangement and protein-protein contact interfaces. The complex was inspected in multiple orientations, with particular attention to the IL-6-gp130 and gp130-sIL-6Rα contact regions that *de novo* binders could sterically target.

(2) PyMOL analysis and definition of functional epitopes

① In PyMOL, three functional epitope regions were defined according to the structural organization of the complex: Part A, corresponding to the IL-6-gp130 interface; Part B, centered on the Trp249-associated gp130-sIL-6Rα interface; and Part C, located around the Trp296 region of sIL-6Rα near the membrane-proximal side. Then we defined binder-specific structural targets as the following.

Table 1. Computational design strategies.

Functional Regions	Structural Target	Key interface/sites	Assigned binders
Part A	IL-6-gp130 interface	IL-6 Leu19/Tyr31 interface	Binder A
Part B	gp130-sIL-6Rα interface	sIL-6Rα Trp249-associated site	Binder B
Part C	sIL-6Rα near-membrane region	sIL-6Rα Trp296-associated site	Binder C

- ② Residues judged to be closely associated with binding at these epitopes were designated as hotspot residues and retained as spatial constraints for subsequent backbone generation.

5. **AI Use Statement:**

Computational prediction generated using AI, not experimentally validated by wet experiment. We employed a combined approach of computational design and experimental verification. Therefore, for results from AI-assisted prediction/simulation, we will continue to conduct wet experimental verification to confirm their validity.